

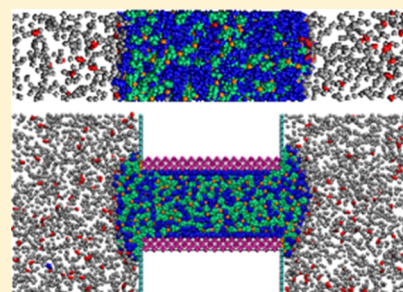
On the Performance of Confined Deep Eutectic Solvents and Ionic Liquids for Separations of Carbon Dioxide from Methane: Molecular Dynamics Simulations

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Supporting Information

ABSTRACT: Classical molecular dynamics simulations were used to investigate the performance of slit graphite and titania (rutile) pores of 5.2 nm in width, partially and completely filled with deep eutectic solvents (DESs) or ionic liquids (ILs), for gas separations of a carbon dioxide–methane mixture of 5:95 molar ratio and temperatures and pressures on the order of 318 K and 100 bar, respectively. The DESs studied were ethaline and levuline (1:2 molar mixtures of choline chloride with ethylene glycol or levulinic acid), and the IL considered was 1-*n*-butyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide, [bmim⁺][NTf₂[−]]. The performance of these systems in terms of solubility selectivity, diffusion selectivity, and permselectivity was compared against the performance of the bulk solvents (which could also be viewed as a model system for the micrometer-sized pores of a supported IL or DES membrane) and against carbon and rutile pores without preadsorbed solvent. The best performance in terms of permselectivity is obtained for bulk levuline and by rutile pores fully filled by ethaline, followed by graphite pores filled by ethaline and the IL. Empty rutile pores have the largest value of solubility selectivity, followed by bulk ethaline and rutile pores completely filled by the IL. The largest values of diffusivity selectivity were observed for bulk levuline, followed by ethaline completely filling a rutile nanopore and a graphite nanopore completely filled with the IL. These observations are rationalized by examining local density profiles and interaction energies among the different entities in our systems. In general, systems of nanopores fully filled by solvents, as well as the bulk solvents, have larger permselectivities than pores partially filled by the IL or the DESs. Drops of 2–3 orders of magnitude are observed in the gas diffusivity in pores filled with solvents with respect to systems of empty pores, which may be problematic if gas permeation is mainly controlled by diffusion. However, if adsorption dominates the gas permeation within the membrane, our results suggest that systems of levuline in the micrometer-sized pores of a supported DES membrane or ethaline confined in the rutile nanopores of a supported DES phase material might represent promising systems for gas separation.



1. INTRODUCTION

Carbon dioxide capture and storage technologies¹ are key in controlling excessive CO₂ emissions from the combustion of fossil fuels. One of the main technologies used worldwide is chemical absorption of CO₂ using solvents such as ammonia or aqueous solutions of amines. However, these approaches have well-known drawbacks such as the energy-intensive process needed to regenerate the absorbent, as well as the corrosive nature and degradability issues of the solvents. As a result, other technologies have been considered to capture carbon dioxide, such as oxyfuel combustion, chemical looping and calcium looping, solid sorbents, ionic liquids (ILs) as alternative solvents, and membrane separations.^{1–3} Replacing the amine-based solvents with ILs attracted a lot of attention over the last few years, with many efforts reported to find suitable ILs (see, e.g., refs^{4–11}). As deep eutectic solvents (DESs) share similar physical and chemical properties as ILs, but are relatively inexpensive and environmentally friendly,^{12–15} they have also been widely studied recently^{16–35} for CO₂ capture applications. Elevated cost is one of the main drawback of ILs, which as DESs typically have high densities and viscosities, leading to mass

transfer issues if one uses these solvents in traditional absorption-based CO₂ separations.^{1,36} These issues can be overcome by incorporating ILs into porous materials, which is the approach used in the development of supported IL membranes (SILMs) and supported IL phase materials (SILPs).^{1,2,4,36–50} This approach has potential advantages such as use of smaller amounts of solvents, good thermal and chemical stabilities, low volatilities, enhanced mechanical strength, and increased contact area between gas and ILs.^{43,45,48–50}

Motivated by the idea of supported IL porous materials for CO₂ capture, in our previous study,⁵¹ we used classical molecular dynamics (MD) simulations to study the adsorption of pure CO₂ gas inside slit-like nanopores with graphite or titania (rutile) walls having different amounts of a typical DES, a 1:2 molar mixture of choline chloride with ethylene glycol (termed ethaline). The average number densities of CO₂ near

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confined ethaline (i.e., dissolved in the DES, adsorbed at the pore walls and at the gas–liquid interface, and present in the gas phase inside the pores) were found to be significantly larger than the CO₂ density within bulk ethaline. In particular, for pores partially filled with DES, the overall average number density of CO₂ could be ~3.0–7.3 times the value observed in bulk ethaline. Our analysis indicated that these observations were driven by interfacial effects and strong interaction energies of CO₂ with ethaline and the pore walls. These results suggested that systems of DESs confined in nanopores could have applications in gas separations. In this work, we again used molecular simulations to investigate how slit-like nanopores with graphitic or rutile walls and containing different amounts of DESs or ILs would perform in the separation of a gas binary mixture of CO₂ and CH₄ with approximate molar ratio of 5:95 and pressures on the order of 100 bar. Such a system is relevant for the separation of carbon dioxide from methane in natural gas. Two different choline chloride-based DESs, namely, 1:2 molar mixtures with ethylene glycol (ethaline) and with levulinic acid (termed levuline) were selected for this study (Figure 1). Both DESs have been studied for CO₂ capture

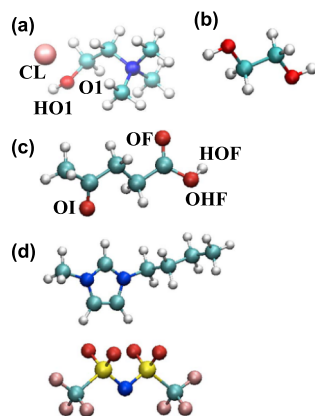


Figure 1. Components of the DESs and IL considered in this work: (a) choline chloride, (b) ethylene glycol, (c) levulinic acid, and (d) [bmim⁺][NTf₂[−]]. Cyan = C, white = H, blue = N, red = O, yellow = S, and pink = Cl (a) or F (d). The labels are atom notations used in the $g(r)$ functions reported in Figure S3 (Supporting Information).

applications in the past.^{17,22,27,29,30,51–53} In addition, we also considered similar systems involving a commonly studied IL, 1-*n*-butyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide, [bmim⁺][NTf₂[−]]^{54–56} for comparison purposes (Figure 1); in particular, Budhathoki et al.⁵⁶ studied the gas separation performance of this IL when confined inside slit graphite pores having widths of 2 and 5 nm. Our goal is to determine the gas separation performance of slit graphite or rutile pores containing varying amounts of these DESs and IL and compare these results with the performance of the bulk DESs and IL. We considered a pore width of 5.2 nm for all our systems, which is similar to the pore size considered by Budhathoki et al.⁵⁶ and is the same pore width we have used in our previous studies of confined ILs^{57–59} and DESs.^{51,60} The gas separation performance of our systems is determined in terms of the solubility and diffusivity selectivities, as well as permselectivities; these variables are defined as follows

$$\beta_{\text{CO}_2, \text{CH}_4}^{\text{S}} = \left(\frac{x_{\text{CO}_2}/y_{\text{CO}_2}}{x_{\text{CH}_4}/y_{\text{CH}_4}} \right)_{\text{T,P}} \quad (1)$$

$$\beta_{\text{CO}_2, \text{CH}_4}^{\text{D}} = \left(\frac{D_{\text{CO}_2}}{D_{\text{CH}_4}} \right)_{\text{T,P}} \quad (2)$$

$$\beta_{\text{CO}_2, \text{CH}_4}^{\text{PI}} = \frac{\Pi_{\text{CO}_2}}{\Pi_{\text{CH}_4}} = \beta_{\text{CO}_2, \text{CH}_4}^{\text{S}} \beta_{\text{CO}_2, \text{CH}_4}^{\text{D}} \quad (3)$$

where x_i and y_i are the mole fractions of the gas component i in the liquid and vapor phase, respectively, D_i is the diffusivity of the gas component i , and Π_i is the permeability of the gas component i . The rest of our paper is organized as follows: Section 2 contains details of the computational models and methods used in this study; our main results are presented and discussed in Section 3, and in Section 4, we summarize the main conclusions of our work.

2. COMPUTATIONAL DETAILS

We performed classical MD simulations of three different DES and IL solvents confined inside slit graphite or TiO₂ [rutile(110)] pores of width $H = 5.2$ nm, in contact with a binary gas mixture of carbon dioxide and methane with an overall molar ratio of 5:95. The two DESs studied consist of choline chloride mixed with either ethylene glycol or levulinic acid, both at a molar ratio of 1:2 (termed ethaline and levuline); the IL system considered in our study was [bmim⁺][NTf₂[−]]. The pore walls used in our systems have lengths of 10.0 and 5.1 nm in the x and y directions, respectively. In addition, we modeled similar bulk systems of DESs and ILs in the center of an elongated box, with the binary gas mixture in contact with both interfaces. Details of our model systems are presented in Table 1, and representative snapshots for our model systems at equilibrium are shown in Figures 2 and S1 (Supporting Information). In Table 1, the name of our model systems is a short-hand notation of solvent (E = ethaline; L = levuline; I = IL), type of system (G = graphite pore; R = rutile pore; B = bulk in elongated box), and number of ion pairs divided by 100. For example, EG3 indicates an ethaline system inside a graphite pore, consisting of 300 choline chloride ion pairs and thus 600 molecules of ethylene glycol as the HBD, in contact with a gas mixture. The pore volume is partially filled with solvent in systems with 300 ion pairs and is completely filled with solvent in systems containing 800 ion pairs (for ethaline, 1000 ion pairs were needed to fully fill a rutile pore so that it forms the same “menisci” protruding into the gas phase as in the levuline and IL systems, Figure 2). Empty graphite and rutile pore systems (G0 and R0) were also considered. We note that we also considered systems with 400 ion pairs; however, the results obtained for these systems were in between the trends observed for systems with 300 ion pairs and pores completely filled with solvents. Therefore, we decided not to include those pore systems with 400 ion pairs in our study for brevity.

The simulation setup for our confined systems is the same used in our previous studies,^{51,60} where the slit pore is in the center of the simulation box and is in contact with two reservoirs at both sides in the x direction (Figures 2 and S1). The whole simulation box has dimensions 30.0, 5.1, and 13.5 nm in the x , y , and z directions, respectively, with auxiliary walls (modeled as graphene layers) placed to prevent the diffusion of

Table 1. Details of Our Model Systems

model system	solvent	wall material	number of ion pairs + HBD molecules	number of carbon dioxide + methane	bulk gas pressure in pore systems (bar) ^a
EG3	ethaline	graphite	300 + 600	210 + 3990	106 ₁
EG8	ethaline	graphite	800 + 1600	180 + 3420	101 ₂
ER3	ethaline	rutile	300 + 600	210 + 3990	103 ₁
ER10	ethaline	rutile	1000 + 2000	180 + 3420	110 ₁
EB	ethaline		450 + 900	90 + 1710	90 ₁
LG3 (u) ^b	levuline	graphite	300 + 600	210 + 3990	112 ₁ ; 111 ₁ (u)
LG8 (u) ^b	levuline	graphite	800 + 1600	180 + 3420	113 ₂ ; 113 ₂ (u)
LR3 (u) ^b	levuline	rutile	300 + 600	210 + 3990	107 ₁ ; 106 ₁ (u)
LR8 (u) ^b	levuline	rutile	800 + 1600	180 + 3420	106 ₁ ; 114 ₁ (u)
LB (u) ^b	levuline		330 + 900	90 + 1710	84 ₁ ; 86 ₁ (u)
IG3	[bmim ⁺][NTf ₂ ⁻]	graphite	300 + 0	210 + 3990	109 ₁
IG8	[bmim ⁺][NTf ₂ ⁻]	graphite	800 + 0	180 + 3420	118 ₂
IR3	[bmim ⁺][NTf ₂ ⁻]	rutile	300 + 0	210 + 3990	101 ₁
IR8	[bmim ⁺][NTf ₂ ⁻]	rutile	800 + 0	180 + 3420	119 ₂
IB	[bmim ⁺][NTf ₂ ⁻]		400 + 0	90 + 1710	
G0		graphite	0 + 0	230 + 4370	106 ₁
R0		rutile	0 + 0	1380 + 4370	111 ₁

^aSubscripts represent uncertainties in the last digit. ^bSystems labeled “u” refer to simulations using the levuline model proposed by Ullah et al.¹⁷

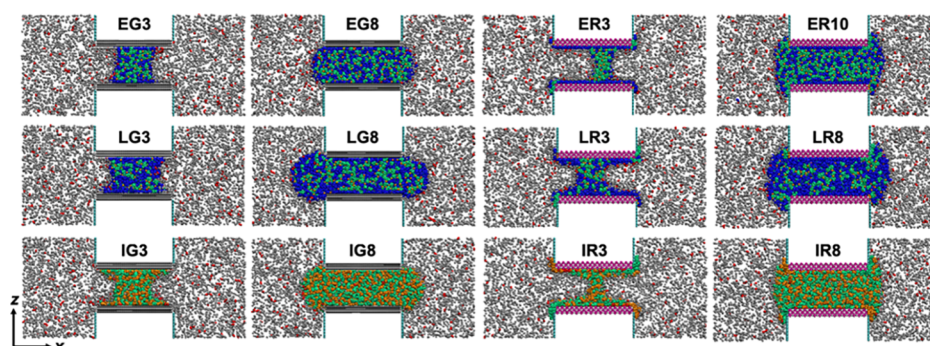


Figure 2. Representative simulation snapshots of pore systems partially or completely filled with solvents at equilibrium at $T = 318$ K (see Table 1 for details). Molecules/atoms are colored as follows: DES/IL: green = cation (choline), orange = anion (chloride), blue = HBD (ethylene glycol/levulinic acid); red = carbon dioxide (CO_2), silver = methane (CH_4). Rutile walls: white = Ti, magenta = O. Graphitic walls: gray = carbon. Auxiliary walls are colored in cyan.

molecules into the vacuum regions that are on the top and bottom of the slit pores in the z direction (Figures 2 and S1). Results presented and discussed in our previous studies^{51,60} indicate that our simulation setup and approach lead to proper equilibration in challenging systems such as those considered here, where complex fluids such as ILs and DESs (solvents that are dense, with strong interactions between their components and slow dynamics) are confined in nanometer-sized pores. As discussed in our previous studies,^{51,60} our rutile pores can suffer from end-wall effects, as one of the edges of the rutile walls has a larger local concentration of O atoms (which have negative partial charges) as compared to the other end, which in turn has a larger amount of Ti atoms (with positive partial charges). As a result, one edge of the rutile walls can have strong interactions with the cations while the other end interacts strongly with the anions, leading in some cases to situations where the edges of the pores are blocked.^{51,60} To reduce these end-wall effects in our rutile systems partially filled with solvent, we initially placed the DES or IL solvents inside the pores using a setup in which the slit pore is not connected to the side reservoirs, that is, the simulation box has the same dimensions in the x and y directions as the slit pore, 10 and 5.1 nm, respectively, and has periodic boundary conditions in those directions. As a result,

the pore walls are infinitely long in the x and y directions, and thus, the rutile walls are free of adsorption “hot spots” at their edges. We pre-equilibrated our pore systems partially filled with DESs or ILs in such a “closed system” setup for 20 ns and then put the pore systems back in the larger simulation box and in contact with the reservoirs containing the binary gas mixtures. Although significant amounts of DES or IL moved to the edges of the rutile walls (Figures 2 and S1), this simulation approach reduced the amounts of DES or IL clustering on the edges of the rutile walls, compared to what was observed in our previous studies.^{51,60} For the rest of our pore systems (graphite pores partially and fully filled with solvent and rutile pores fully filled with solvent), we used the same procedure as in our previous works,^{51,60} that is, initially placed all species in the side reservoirs and allowed them to diffuse into the pore. The simulation box in our elongated bulk systems has dimensions $4.5 \text{ nm} \times 4.5 \text{ nm} \times 40 \text{ nm}$ in the x , y , and z directions (Figure S1); these systems contain different amounts of DES or IL, as these solvents have different densities. All our simulations were performed at a temperature $T = 318$ K using GROMACS 5.1.4.^{61–67} The initial position of the atoms in choline chloride, ethylene glycol, levulinic acid, and [bmim⁺][NTf₂⁻] was obtained using PRODRG Server.⁶⁸

The force field parameters used for the rutile and graphite walls in our systems are the same used in our previous studies.^{51,60} Carbon dioxide and methane molecules were modeled using the TraPPE force field.⁶⁹ The force field and parameters used for [bmim⁺][NTf₂⁻] were taken from the work of Budhathoki et al.,⁵⁶ whereas those used for choline chloride and ethylene glycol were obtained from the study by Perkins and co-workers;⁷⁰ both of these models are based on the general AMBER force field (GAFF).⁷¹ Two different models were used for levuline. Our first model was developed following the methodology used by Perkins et al.^{70,72} We first obtained the optimized structure of levulinic acid using Gaussian 16⁷³ at the HF/6-31G* level of theory and then determined partial charges using the restrained electrostatic potential charge derivation method from the AMBER tools.^{74,75} Finally, the partial charges of the atoms in the ions were scaled as to yield a total ion charge of $\pm 0.8e$. Such a model for levuline is consistent with the ethaline model used here and with all DES models used in our previous studies.^{51,60,76} The second model considered here was proposed by Ullah et al.¹⁷ (systems labeled in Table 1 using “u”, e.g., LBU). Instead of using $\pm 1e$ for the ions and zero charge for neutral molecules, Ullah et al. developed their parameterization using DFT calculations for optimized clusters formed by a choline chloride ion pair and two molecules of levulinic acid, yielding charges of $+0.8254e$ for choline, $-0.6849e$ for chlorine, and $-0.0663e$ and $-0.0743e$ for the two molecules of levulinic acid.^{17,52} This levuline model yields density values that are in good agreement with experimental data.^{17,53} We note that both levuline models used here give density and self-diffusion coefficient values that were similar to each other. Nevertheless, both levuline models exhibit differences in the radial distribution functions (RDFs) $g(r)$, as determined from separate NVT simulations of levuline at 318 K (250 ion pairs and 500 molecules of levulinic acid), which yielded slightly different densities for both models: $1108 \pm 4 \text{ kg/m}^3$ (GAFF-based model) and $1133 \pm 4 \text{ kg/m}^3$ (model by Ullah et al.¹⁷) The RDFs between the center of mass (COM) of the ions and levulinic acid, as well as between main atoms in levuline species are presented in Figures S2 and S3, and differences are discussed in the Supporting Information. Nevertheless, and despite the differences observed in the liquid structure, similar gas separation results were observed when using both levuline models.

Packmol⁷⁷ was used to randomly place all molecules (DES, IL, gas) in our initial configurations, which were then subjected to an energy minimization procedure using the steepest descent method. We then ran short MD simulations at $T = 500 \text{ K}$ for 5 ns and then annealed to 318 K over 2 ns. Our model systems were then initially equilibrated at this temperature for 5 ns, followed by a 100 ns-long production run. Finally, the above steps (starting from running at $T = 500 \text{ K}$) were repeated starting from the configuration obtained at the end of the first production run. Therefore, for most of our systems, we gathered two independent simulation runs for each system described in Table 1, and our results were averaged over these two independent realizations of each system. The exception were the pore systems fully filled with solvent, for which four independent realizations of each system were required to obtain reasonably good statistics, especially for the compositions and diffusion coefficients of CO₂ and CH₄. As done in our previous study,⁵¹ we estimated the pressure of the bulk gas in contact with our pore systems by conducting separate MD simulations in the NVT ensemble of the bulk gas at the same temperature, T

= 318 K, using densities and compositions equal to the average values observed in the reservoirs at both sides of our slit pores. The computed pressures are reported in Table 1 and are similar to those considered by Budhathoki et al. in their study.⁵⁶

3. RESULTS AND DISCUSSION

3.1. Gas Separation Performance of Studied Systems.

In Figure 3 we present the computed solubility selectivities of

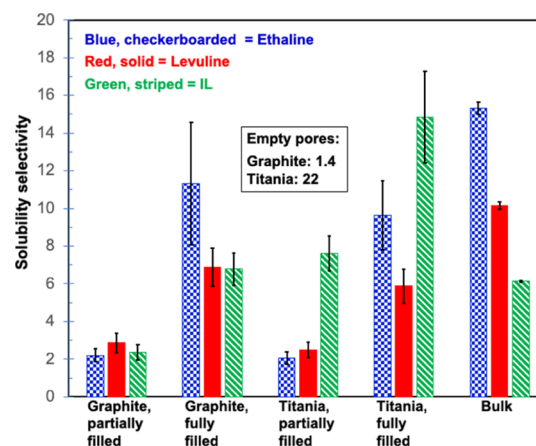


Figure 3. CO₂/CH₄ solubility selectivity in ethaline, levuline or IL systems partially or completely filling slit graphite and titania nanopores, and in the bulk. Solubility selectivities for graphite and rutile pores with no preadsorbed solvent are also reported. All relevant data are presented in Table S1 (Supporting Information).

CO₂ over CH₄ for slit graphite and titania nanopores partially and completely filled with ethaline, levuline, and the IL, which represent our model-supported IL (or DES) phase materials. Figure 3 also depicts results obtained for the bulk solvents, which can be seen as the solubility selectivity obtained in the micrometer-sized pores of a supported IL (or DES) membrane. Following eq 1, the solubility selectivity is calculated in bulk systems by taking the ratio of the mole fractions of CO₂ in the liquid solvent and in the gas phase and then dividing by the ratio of mole fractions of CH₄ in the liquid and gas. In our nanopore systems, y represents the mole fraction of CO₂ or CH₄ in the bulk gas phase (i.e., outside of the pore) and x represents the mole fraction of CO₂ or CH₄ in the whole pore volume, which is calculated by dividing the number of CO₂ or CH₄ molecules by the total number of ions and molecules inside the pore. Therefore, for pores fully filled with solvent, the mole fractions x accounts for gas molecules dissolved in the confined solvent and adsorbed at the pore walls; as the gas/solvent interfaces in these systems are typically outside of the pore volume (Figure 2), we did not consider the CO₂ or CH₄ molecules adsorbed at those interfaces when computing x . For the case of pores partially filled with DESs or IL, x accounts for the gas molecules retained at the gas/solvent interface, as well as those molecules present in the gas phase inside the pore, in addition to the gas molecules dissolved in the confined solvent and adsorbed at the pore walls (Figure 2). All the data used to compute the solubility selectivities for all our systems are included in Table S1 (Supporting Information). The results shown in Figure 3 and Table S1 indicate that the systems of bulk ethaline and the slit titania nanopore completely filled with the IL have the largest solubility selectivities of CO₂ over CH₄ (~ 15 in both cases), followed by the slit graphite nanopore

filled with ethaline (~11) and bulk levuline (~10.1 when using the GAFF-based model, ~8.2 from the model by Ullah et al.,¹⁷ Table S1) and the titania nanopore filled with ethaline (~10). Nevertheless, the solubility selectivity computed for a titania nanopore with no DES or IL present is the highest among all systems evaluated, ~22 (Figure 3 and Table S1), which suggests that a simple slit rutile nanopore would be better for adsorption-based separations of CO₂ from CH₄. An empty graphite pore had the lowest value of solubility selectivity among all systems studied here (~1.4). These observations are not surprising, as CO₂ adsorbs strongly to the rutile walls in titania pores without preadsorbed solvent, whereas CH₄ is preferentially adsorbed by the carbon walls over CO₂ in graphite pores with no preadsorbed solvent (see Figures 6 and S1, and Section 3.2 below).

In our previous study⁵¹ we found that slit graphite and rutile pores partially filled with ethaline retained important amounts of CO₂ at the DES/gas interface, at the pore walls, and dissolved within the nanoconfined DES. However, the results shown in Figure 3 and Table S1 indicate that in general, slit graphite or titania pores completely filled with DES or IL outperform similar pores partially filled with the same solvent in terms of solubility selectivities. In addition, confinement inside the nanopores studied here does not seem to lead to significant improvements in the solubility selectivities for ethaline and levuline, as all nanoconfined DES systems had lower solubility selectivities compared to the same DES in the bulk. In contrast, graphite and specially rutile nanopores completely filled with the IL outperform the bulk IL in terms of solubility selectivity; rutile pores partially filled with the IL also have slightly higher solubility selectivities compared to the bulk IL. These observations for the IL can be explained by the strong interactions between carbon dioxide and the rutile walls in pores partially and completely filled with the IL (Figure 8 and Section 3.3 below). Discussing the confinement effect on gas solubilities, the results shown in Table S1 (Supporting Information) indicate that gas solubility is affected by the amount of IL or DES solvent in the pores. In general, the solubility of both CO₂ and CH₄ in pores partially filled with solvent is larger compared to the gas solubilities observed in the bulk solvent, as significant amounts of gas molecules are retained at the gas/solvent interface, dissolved in the confined solvent, adsorbed at the pore walls, and present in the gas phase inside the pore. However, the solubility selectivity in pores partially filled with all solvents is in general lower compared to the values observed for the bulk solvents, as the solubility increases for both CO₂ and CH₄ in these systems; the only exception is the IL partially filling a rutile pore, which has a slightly larger solubility selectivity compared to the bulk IL. For pores fully filled with solvent, in general, the solubility of CO₂ in both DESs is very similar to the values observed in the bulk DESs. The solubility of CH₄ in both confined DESs is in general larger compared to the values obtained for the bulk DESs; for ethaline, slightly larger values are observed in the rutile pore as compared to the graphite pore, whereas for levuline, slightly larger CH₄ solubilities are found in the graphite pore compared to the titania pore. For pores fully filled with IL, graphitic pores show CO₂ and CH₄ solubilities that are, respectively, similar and smaller compared to the values observed in the bulk IL, whereas rutile pores have CO₂ and CH₄ solubilities that are, respectively, larger and smaller compared to the values observed in the bulk IL, as CO₂ interacts strongly with the rutile walls in IL systems (Figure 8). As a result, the ILs partially and

completely filling a rutile pore are the only systems that show larger solubility selectivities compared to the bulk solvent.

We note that in our simulations, the bulk gas pressure is not exactly the same in all our systems (Table 1). The gas pressures in our bulk systems are lower (~84–90 bar) than the values considered in our pore systems (~101–118 bar); however, we do not anticipate that our results and trends for bulk systems would change significantly if larger values of gas pressures were considered for our bulk systems. We also note that we started our simulations with a fixed number of molecules of CO₂ and CH₄, with an initial molar ratio of 5:95; therefore, larger amounts of CO₂ molecules inside the pore translates into smaller amounts of carbon dioxide in the bulk gas phase. One way to overcome these issues is to perform Monte Carlo (MC) simulations in the grand canonical ensemble (GCMC) or isothermal–isobaric Gibbs ensemble MC (GEMC) simulations, in which the pore systems are by design in equilibrium with a bulk gas phase. In GCMC simulations, the fact that the chemical potentials of CO₂ and CH₄ in the gas phase are the same as the values inside the pore translates into a bulk gas phase with fixed pressure and composition. However, special simulation techniques would be needed to increase the success rate of trial moves where gas molecules are inserted or removed from the confined solvents, which in the case of DESs or ILs are dense and viscous. Nevertheless, we do note that the solubility selectivities computed from our MD simulations for the IL completely filling a slit graphite nanopore (6.8 ± 0.9) and in the bulk (6.1 ± 0.1) are in general agreement with the values reported by Budhathoki et al.⁵⁶ for the same systems (8.7 ± 0.4 and 6.2 ± 0.8 , respectively), which were computed from rigorous calculations of the gas solubilities using GEMC simulations.

In Figure 4, we present results for the diffusion selectivities of CO₂ over CH₄ for slit graphite and titania nanopores partially

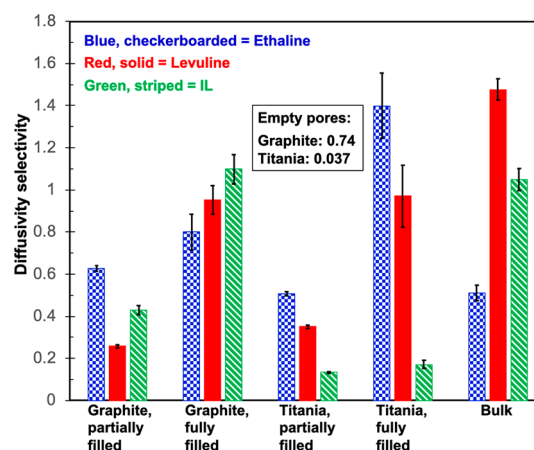


Figure 4. CO₂/CH₄ diffusivity selectivity in ethaline, levuline or IL systems partially or completely filling slit graphite and titania nanopores, and in the bulk. Diffusivity selectivities for graphite and rutile pores with no preadsorbed solvent is also reported. All relevant data are presented in Table S2 (Supporting Information).

and completely filled with ethaline, levuline and the IL, and for the bulk solvents. These results were computed according to eq 2 using the diffusion coefficients determined for CO₂ and CH₄ within the different pore systems. All the data used to compute the diffusivity selectivities for all our systems are included in Table S2 (Supporting Information). The diffusion coefficient of

any given species is calculated from its measured mean-squared displacement, using the relevant Einstein relation in three dimensions; our pore size ($H = 5.2$ nm) is not narrow enough for our systems to be strictly considered as two-dimensional. Displacements of all relevant molecules/ions (see below) contribute to a single average for each molecule/ion type. For the case of pore systems, we only considered the gas molecules within the nanopores; in particular, for pores fully filled with solvent, we did not consider the gas molecules retained in the gas–liquid interface in the “menisci” protruding into the side reservoirs (see Figure 2). In the case of bulk systems in elongated boxes, we only considered the gas molecules dissolved within the solvents (i.e., away from the gas–liquid interface); we arbitrarily restricted our analysis to those gas molecules specifically located in the region delimited by the first local minima in the number density profiles of CH_4 within the liquid solvent (Figure 6, systems EB, LB and IB, and Figure S4); these local minima are typically about 1 nm away from the CH_4 density peaks. As noted in Section 2 above, for the case of pores fully filled with solvent, we averaged our gas diffusivity results over four independent simulation runs to improve the accuracy of our calculations, but we still observed significant variations in the diffusivities and selectivities calculated in these systems (Figure 4 and Table S2). To further confirm our diffusivity results, we also conducted additional simulations for pores fully filled with solvent in the NVT ensemble, considering just the pore volume (i.e., no interfaces or gas reservoirs at the side of the pore). The number of molecules of all species (CO_2 , CH_4 , cations, anions, and HBD) considered in these additional simulations was chosen to be similar to the average number of molecules computed from our original MD simulations. Likewise, for bulk solvent systems, we corroborated our diffusivity results by performing additional simulations in the NVT ensemble, considering just the liquid phase (i.e., no interfaces or gas phase); the total volume and the number of molecules of all species in these simulations were chosen to yield similar number densities determined in the liquid phase in the elongated box systems. A final observation related to the results shown in Figure 4 and Table S2 is that in pores partially filled with solvents, we computed the diffusivity selectivities using the “apparent” diffusion coefficients of the gas molecules, as determined by considering all CO_2 or CH_4 molecules inside the pore (i.e., dissolved in the solvent, adsorbed at interfaces or the pore walls, and in the gas phase in the pore). Naturally, the CO_2 and CH_4 molecules in the gas phase inside the pore move much faster than those gas molecules adsorbed at the gas–liquid interface or at the pore walls or dissolved within the solvent. In consequence, the apparent diffusivities for CO_2 and CH_4 become larger as the amounts of DESs or IL inside the pores decrease (Table S2).

From Figure 4 and Table S2, the largest diffusivity selectivities were observed for bulk levuline (~ 1.48 from the GAFF-based model; ~ 1.7 from the model by Ullah et al.¹⁷), ethaline completely filling a slit titania pore (~ 1.4), and a graphite pore completely filled by the IL (~ 1.1 , in good agreement with the value reported by Budhathoki et al.,⁵⁶ ~ 1.2). In general, ethaline systems in the bulk or inside slit graphite pores show diffusion selectivities lower than 1; however, confinement of ethaline in a slit rutile pore causes carbon dioxide to become more mobile than methane and the diffusivity selectivity jumps to ~ 1.4 . CO_2 interacts strongly with ethylene glycol in bulk ethaline (Figure 8). However, when ethaline is confined inside rutile pores, the HBD is strongly

adsorbed by the titania walls (Figure 7), causing the local density of ethylene glycol to drop farther away from the pore walls, which might help increase the diffusivity of CO_2 relative to that of CH_4 . In contrast, bulk levuline has a high diffusivity selectivity (~ 1.5 – 1.7), but confinement inside graphite or rutile nanopores causes the diffusivity selectivity to drop. CO_2 also interacts very strongly with levulinic acid in bulk levuline (Figure 8), but this HBD is adsorbed less strongly by the rutile and graphite walls compared to ethaline (Figure 7), which might explain the fact that confinement does not seem to help gas separations in terms of diffusivity selectivity for levuline systems. For the IL, confinement inside a slit graphite nanopore causes the diffusivity selectivity of CO_2 over CH_4 to remain almost unaltered (~ 1.1) with respect to the value observed for the bulk IL (~ 1.05 , also in excellent agreement with the value reported by Budhathoki et al.,⁵⁶ ~ 1.05). An empty rutile nanopore, which outperformed all studied systems in terms of solubility selectivity (Figure 3), has a much lower diffusivity selectivity compared to the rest of our systems, as CO_2 is strongly adsorbed by the titania walls (Figure 6) and causes its diffusivity to be much smaller than that of CH_4 . Likewise, a graphite pore with no preadsorbed DES or IL has lower diffusivity selectivity than the same pore completely filled with any of the solvents considered here.

The results shown in Figure 4 and Table S2 indicate that slit graphite or titania nanopores fully filled with DES or IL outperform similar pores partially filled with the same solvent in terms of diffusivity selectivities. Comparing gas diffusivity values in our systems, we observe drops of up to 3 orders of magnitude in the values determined in pores fully filled with solvent, with respect to the values determined in empty pores; smaller drops in gas diffusivities are observed in pores partially filled with solvent. For rutile pores, however, an increase in CO_2 diffusivity is observed in pores partially filled with DES, as the HBD adsorbs strongly to the titania walls (see Figure 7 below) and displaces carbon dioxide into the gas phase, increasing its mobility; filling the pores with solvent again lead to a decrease in diffusivity for both carbon dioxide and methane. Comparing gas diffusivities in ethaline and levuline systems in the bulk and in fully filled nanopores, we observe slight increases in CO_2 diffusivities in rutile pores completely filled with the DESs compared to the values observed in the bulk solvents, which can be caused by the strong adsorption of the HBD (especially ethylene glycol) by the titania walls (Figure 7). For graphite pores, the changes in gas diffusivity with respect to the values determined for the bulk solvents are modest and seem to depend on the system, as reductions are observed for the case of ethaline and almost no changes are observed for levuline. For the case of the IL, the gas diffusivities in the bulk IL are very similar to the values determined in filled graphite pores, but in rutile pores fully filled by IL, the CO_2 diffusivity is smaller and the CH_4 diffusivity is larger than the values observed in the bulk IL.

To discuss effects of confinement on the diffusivities of the components of the solvents, in Table S3 (Supporting Information), we report the diffusion coefficients for the cations, anions, and HBD species in all our model systems. These results indicate that confinement effects seem to depend strongly on the specific system. For ethaline systems, in general, confinement and strong interactions with the pore walls, especially in the case of rutile pores, tend to reduce the diffusivities of the cations, anions, and HBD with respect to the bulk values; however, diffusivities in system EG3 are larger than

the bulk values, as the pore is not completely filled with ethaline and the interactions with the graphitic walls in this system seem to be smaller compared to the interactions with the walls in the rest of the systems (Figure 8). In contrast, in systems containing levuline or $[\text{bmim}^+][\text{NTf}_2^-]$, in general, all species have larger diffusivities when inside the nanopores compared to the values observed in bulk systems, with the exception of the levuline and IL cations, and the IL anions, inside partially filled rutile pores, which move slower compared to their counterparts in bulk systems. In addition, the anions in systems LR3u also have smaller diffusivities compared to the values observed for the anions in system LBU. We note that all our systems contain CO_2 and CH_4 , which tend to increase the diffusivities of the solvent components with respect to similar systems that contain no gases.⁵¹ Finally, we note that in previous studies, we have compared the dynamics of ILs^{57,59} and DESs⁶⁰ confined in slit graphite and rutile pores to the mobilities of the bulk solvents, although these ILs and DESs were different from the solvents considered here.

In Figure 5 we present results for the permselectivities of CO_2 over CH_4 for slit graphite and titania nanopores partially

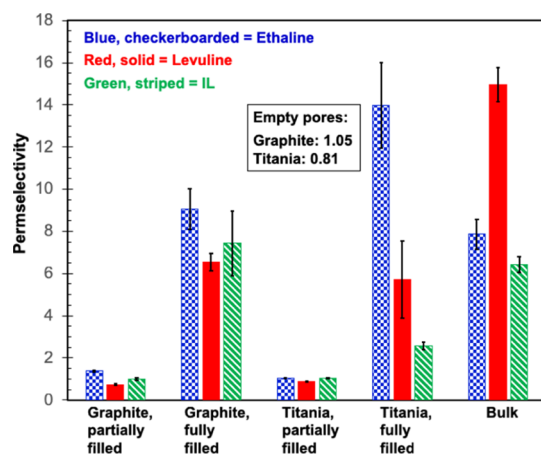


Figure 5. CO_2/CH_4 permselectivity in ethaline, levuline or IL systems partially or completely filling slit graphite and titania nanopores, and in the bulk. Permselectivities for graphite and rutile pores with no preadsorbed solvent is also reported. All relevant data are presented in Table S2 (Supporting Information).

and completely filled with ethaline, levuline and the IL, and for the bulk solvents. Permselectivities were calculated as the product of the solubility and diffusivity selectivities, according to eq 3. All relevant data are included in Table S2 (Supporting Information). Bulk levuline shows the largest value of permselectivity among our systems (~ 15 from both force fields considered), followed by rutile nanopores completely filled with ethaline (~ 14), ethaline completely filling a graphite pore (~ 9), or in the bulk (~ 7.9). The permselectivity computed for a graphite pore fully filled with the IL was 7 ± 2 , which is in general agreement with the value determined by Budhathoki et al.,⁵⁶ 10 ± 1 ; discrepancies are mainly due to differences in the computed solubility selectivities as discussed above. We do note, however, that the permselectivity computed by us for the bulk IL, 6.4 ± 0.4 , is in excellent agreement with the value reported by Budhathoki et al.,⁵⁶ 6.4 ± 0.9 . The results shown in Figure 5 and Table S2 indicate that nanopores partially filled with solvents have much lower permselectivities compared to completely filled pores. Although our previous study⁵¹

suggested that important amounts of CO_2 were retained in pores partially filled with DES at the solvent/gas interfaces, at the pore walls, and dissolved within the confined DES, the results shown in Figure 5 and Table S2 indicate that these interfacial effects do not translate into satisfactory gas separation capabilities. When comparing the performance of pores fully filled with solvent against the bulk systems in terms of permselectivities, in the case of ethaline systems, our results suggest that confinement in slit graphite or rutile nanopores lead to up to twofold increases in the permselectivity with respect to the values observed for bulk ethaline. These observations are driven by the strong adsorption of ethylene glycol by the pore walls, especially in the case of rutile pores (Figures 6 and 7), which lead to depletion of the HBD near the

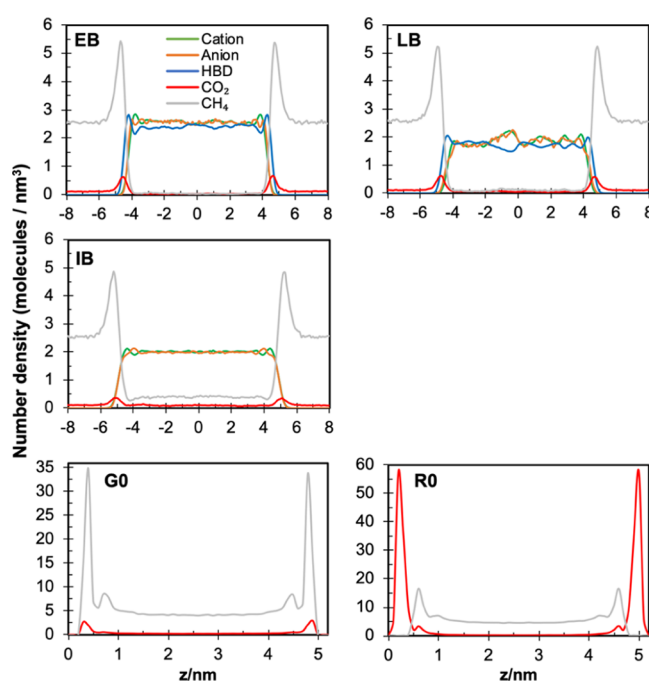


Figure 6. Number density profiles of DES/IL species and gas molecules along the z -direction for bulk systems in elongated boxes (systems EB, LB, and IB, Table 1) and graphite/rutile nanopores with no preadsorbed solvent (systems G0 and R0, Table 1). For ease of visualization, the number density of HBD shown for DES systems is the actual density divided by 2 (as the ratio between HBD and the ions is 2). The density profiles were computed based on the positions of the COM of any given molecule/ion.

center of the pore and might help increase the mobility of CO_2 with respect to CH_4 . In contrast, confinement leads to reduced permselectivity values for levuline compared to its bulk value, and for the IL, confinement in slit graphite pores leads to a slight increase in permselectivity, whereas confinement in a titania pore leads to a reduction in permselectivity compared to the value determined for the bulk IL. All systems of nanopores filled with solvents, as well as the bulk solvent systems, outperform graphite or rutile nanopores with no preadsorbed DESs or IL in terms of permselectivities. However, as noted above and by Budhathoki et al.,⁵⁶ drops of 2–3 orders of magnitude are observed in the gas diffusivity in pores filled with solvents with respect to empty pores, which may pose a problem if gas diffusion within the pores is the main phenomena controlling permeation in the membrane. However, if adsorption is the controlling step in gas permeation

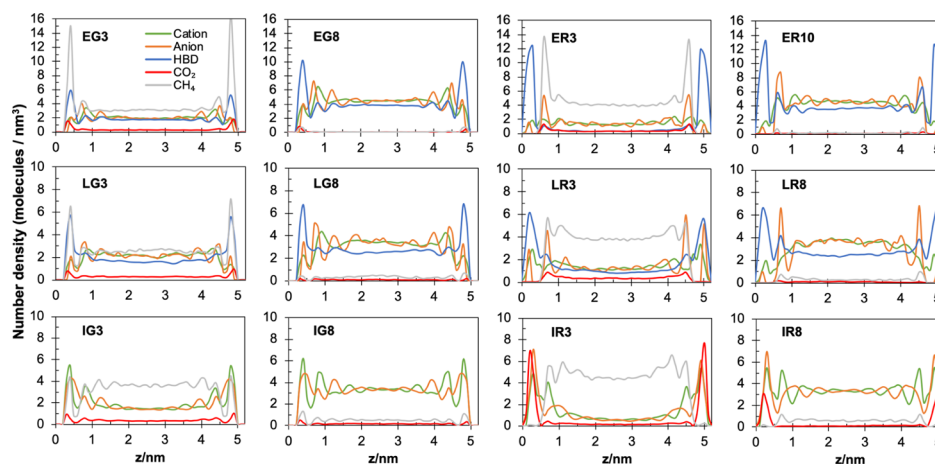


Figure 7. Number density profiles of DES/IL species and gas molecules along the z -direction for all pore systems (see Table 1 for details of model systems). For ease of visualization, the number density of HBD shown for DES systems is the actual density divided by 2 (as the ratio between HBD and the ions is 2). The density profiles were computed based on the positions of the COM of any given molecule/ion.

within the membrane, then, systems of ethaline confined in rutile nanopores, levuline in the micrometer-sized pores of a supported DES membrane, or ethaline or the IL confined in graphite nanopores might represent promising approaches for gas separation. However, we note that high pressure differences⁷⁸ and solvent leaching⁷⁹ are typical problems in SILMs and SILPs; solvent leaching becomes more problematic if an expensive solvent is used. Even though the DESs considered here are not expensive, one might need to consider approaches such as covalent binding⁷⁹ of solvent components to the pore walls.

Even though our equilibrium simulations do not provide direct results for permeability because they are conducted without a driving force for separation, results from our simulations can be used to provide gas permeability estimates. Here, we assume that the gas retained in our porous matrices (i.e., our graphite and rutile nanopores and considering that the bulk solvents can be viewed as model systems for the micrometer-sized pores of a supported IL or DES membrane) is in equilibrium with the bulk gas reservoir, which is at pressures on the order of 100 bar and thus departs from ideal gas behavior. The amount of gas retained by the porous matrix can be expressed in terms of the volume that the gas would occupy at standard conditions, for which ideal gas behavior would be expected. Following previous studies,^{80,81} we obtain the following equation for the gas solubility (CO_2 or CH_4) in our porous matrices

$$S_{\text{gas}} = \frac{V_{\text{gas}}(\text{STP})/V_{\text{m}}}{\hat{f}_{\text{gas}}^{\text{V}}} = \frac{N_{\text{gas}}k_{\text{B}}T_{\text{S}}}{P_{\text{S}}V_{\text{m}}\hat{f}_{\text{gas}}^{\text{V}}} \quad (4)$$

where $V_{\text{gas}}(\text{STP})$ is the volume occupied by N_{gas} molecules (CO_2 or CH_4) at standard conditions ($T_{\text{S}} = 273.15 \text{ K}$, $P_{\text{S}} = 1.013 \text{ bar}$). The N_{gas} molecules are adsorbed in a porous membrane of volume V_{m} , k_{B} is the Boltzmann constant, and $\hat{f}_{\text{gas}}^{\text{V}}$ represents the fugacity of CO_2 or CH_4 in the bulk gas reservoir. This last variable was determined using the Peng–Robinson equation of state with $k_{ij} = 0.092$,^{82,83} using as input the pressure and the composition of the bulk gas (Tables 1 and S1 respectively) at the temperature of our systems, $T = 318 \text{ K}$. Equation 4 yields the gas solubility in units of $(\text{cm}^3 \text{ of gas at STP})/[(\text{cm}^3 \text{ of membrane}) \times (\text{bar})]$, which can then be multiplied by the gas diffusivity (in cm^2/s) within the

membrane to yield the gas permeability, which is usually reported in barrers [$1 \text{ barrer} = 10^{-10} \text{ cm}^3 (\text{STP}) \text{ cm cm}^{-2} \text{ s}^{-2} (\text{cm Hg})^{-1}$]

$$\Pi_{\text{gas}} = S_{\text{gas}} D_{\text{gas}} \quad (5)$$

Table S4 (Supporting Information) includes results for the gas solubilities and permeabilities computed through eqs 4 and 5, as well as the permselectivities calculated as the ratio of the gas permeabilities. We first note that the permselectivities shown in Table S4 are in general slightly larger than the permselectivities reported in Figure 5 and Table S2 (which were computed as the product of the solubility and diffusivity selectivities determined from simulation data). This observation might be caused by uncertainties in N_{gas} and $\hat{f}_{\text{gas}}^{\text{V}}$ (which in turn is affected by uncertainties in the pressure and composition of the bulk gas). Nevertheless, the permselectivity trends from the values reported in Table S4 are consistent with the results presented in Figure 5 and Table S2, and thus, our analysis and conclusions remain unaltered. Although our permeability results exhibit variations of up to 5 orders of magnitude among our systems, our computed values are consistent with experimental CO_2 permeabilities measured in SILMs and SILPs (see, e.g.,^{46,84,85} and references therein). In general, the largest gas permeabilities are observed for pores without any preadsorbed solvent ($\sim 10^5$ barrers), followed by pores partially filled with solvents ($\sim 10^4$ barrers). The bulk solvents (which are representative of the micron-sized pores in supported IL or DES membranes) and the nanopores fully filled with solvent have permeabilities on the order of $\sim 10^2$ to 10^0 barrers. Among the bulk solvents, the IL has the largest CO_2 permeability, followed by bulk levuline and bulk ethaline; the IL also has the largest CH_4 permeability, followed by bulk ethaline and levuline. Focusing now on the bulk solvents and the fully filled pores, which exhibit the largest permselectivities, we note that in general, the IL systems have the largest CO_2 and CH_4 permeabilities, with the bulk IL having the largest CO_2 permeability followed by the IL completely filling the graphite and rutile pores. These trends indicate that for IL systems, confinement seems to lead to reductions in the CO_2 permeability with respect to the value determined for the bulk IL. Similarly, the largest CH_4 permeability is observed for the IL filling a rutile pore, followed by the bulk IL and the IL inside a graphite pore. Focusing now on the DES systems in the

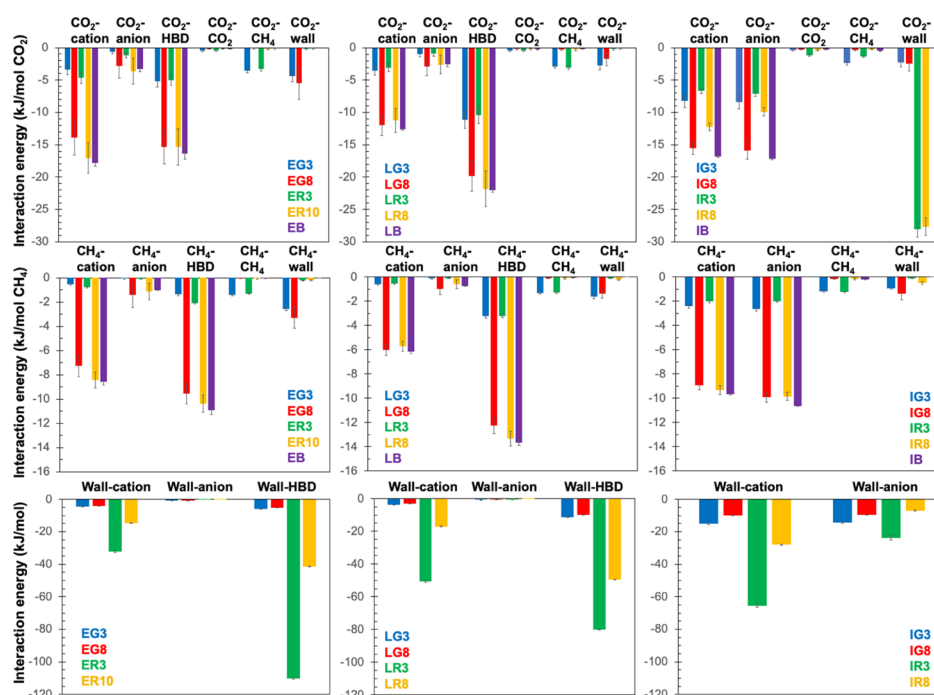


Figure 8. Average interaction energies (electrostatic + van der Waals) in pore and bulk systems. Top row: CO_2 -species (in kJ per mol of CO_2); middle row: CH_4 -species (in kJ per mol of CH_4); bottom row: wall-species (in kJ per mol of cation, anion or HBD). Left column: Ethaline systems; center column: levuline systems; right column: IL systems. All data are reported in Tables S5–S7 (Supporting Information).

bulk and fully filling nanopores, confining ethaline or levuline in a graphitic pore leads to drops in CO_2 and CH_4 permeability with respect to the value observed in the bulk DESs, whereas completely filling a rutile pore with any of the DESs leads to a twofold increase in CO_2 permeability with respect to the value determined for the bulk DES. Similarly, the CH_4 permeability for levuline filling a rutile pore is about 7 times the value observed for bulk levuline. These observations are related to the strong interactions between the DES components (mainly the HBD) with the rutile walls (Figure 8). As reflected in the computed permselectivities for the bulk solvents and for nanopores completely filled by solvents, the CO_2 permeabilities are in general larger than the values observed for CH_4 . Pores partially filled with solvent in general have the lowest permselectivity values among our systems, and for pores partially filled with levuline, the CO_2 permeabilities are smaller than those of methane.

3.2. Local Density Profiles. In Figure 6, we present the local density profiles of all species in our bulk systems in elongated simulation boxes and in slit graphite and rutile pores without any preadsorbed solvents; similar density profiles for bulk levuline using the model by Ullah et al.¹⁷ are shown in Figure S4 (Supporting Information). As results from our previous study⁵¹ indicated that important amounts of carbon dioxide adsorb at the gas–DES interface, here, we started by examining density profiles for CO_2 , CH_4 , and each of the components of the DESs or IL as a function of the z -coordinate (Figure 6) in bulk solvent systems placed in the center of an elongated simulation box, thus creating two gas–liquid interfaces (Figure S1). In all bulk DES systems, the density peak corresponding to the HBD (ethylene glycol or levulinic acid) is closer to the gas side of the interface as compared to the density peaks of the cation or anion, indicating that the HBD is the predominant species in the liquid side of the gas/solvent interface. These trends are similar to those reported in our

previous simulations⁵¹ for bulk ethaline in contact with a gas phase of pure CO_2 . The density profiles obtained for bulk levuline using both models are in general agreement (Figures 5 and S4), with slight differences observed mainly in the density peak of the HBD at the interface, and in the average values of the densities of HBD, cation and anion in the center of the DES slab away from the interface; similar gas densities were obtained for levuline using both models considered here. For the bulk IL system, the density peak of the cation appears closer to the gas side of the interface than the anion's density peak. CO_2 and CH_4 also have large density peaks at the gas–liquid interface, but in all bulk systems examined in Figure 6, CH_4 prefers to lie closer to the gas side of the interface, whereas the density peak of CO_2 appears closer to the liquid side of the interface. In DES systems, the density peak of CO_2 lies close to the density peak of the HBD at the interface, whereas in the IL system, the density peak of carbon dioxide is closer to the cation's density peak. Examination of the average number densities of CO_2 and CH_4 determined in the bulk of the liquid solvents, as well as the x_i values reported in Table S1 for systems EB, LB, LBU, and IB, suggests that the IL can solubilize the largest amounts of CO_2 and CH_4 in its bulk liquid phase, followed by levuline and ethaline. The results presented in Figure 6 also show large local densities of CO_2 at the gas–liquid interface for all bulk solvent systems. The largest values of CO_2 density at the interface, as well as the largest differences between the CO_2 density at the interface and in the bulk liquid, are observed for ethaline, followed by levuline and lastly the IL, with similar trends observed for CH_4 . The density profiles of CO_2 and CH_4 in graphite and rutile pores with no preadsorbed solvent (Figure 6; see also Figure S1) indicate that large quantities of carbon dioxide are adsorbed by the rutile walls, with a layer of methane containing smaller amounts of CO_2 adsorbed next to this carbon dioxide layer. In contrast, a graphite pore without any

solvent mainly adsorbs CH_4 , with a small amount of CO_2 adsorbed next to the carbon walls.

The local number density profiles of CO_2 , CH_4 , and each of the components of the DESs or IL as a function of the z -coordinate inside our pore systems are presented in Figure 7; similar density profiles for confined levuline using the model by Ullah et al.¹⁷ are shown in Figure S5 (Supporting Information). As found in our previous studies,^{51,60} among the components of the DES, the HBD species exhibit the largest local densities in the layers of fluid that are close to the pore surfaces in both graphite and rutile systems. Comparing the two HBDs, levulinic acid has smaller densities near the rutile walls as compared to ethylene glycol, as the latter has two OH groups that can form hydrogen bonds with the oxygen atoms in the rutile walls while levulinic acid only has one hydroxyl group (however, the O atoms from levulinic acid can interact strongly with the Ti atoms in the walls). The smaller density of levulinic acid near the rutile walls leads to larger densities for the ions near the rutile walls for levuline as compared to ethaline, where ethylene glycol covers most of the rutile surfaces. Comparing systems of graphite pores partially filled with DES, we observe that levulinic acid has larger densities near the graphite walls as compared to ethylene glycol, but in graphite pores fully filled with DESs, ethylene glycol has a larger density peak than levulinic acid near the pore walls. The density profiles obtained for the levuline components using the two different force fields show small but noticeable differences, namely, (1) density profiles for the ions and HBD in the first and second layers that are qualitatively and quantitatively different and (2) density profiles for the ions in the center region of the pore tend to be smoother with the GAFF-based model as compared to the model by Ullah et al.,¹⁷ with small but noticeable differences in the average density values for all DES components between the two different levuline models. These observations are caused by the fact that the two force fields yield slightly different liquid structures for levuline (Figures S3 and S4). Now focusing on the IL systems, the anions tend to have slightly larger density peaks near the rutile walls as compared to the cations, whereas near graphite walls, the cations tend to have slightly taller density peaks than the anions.

We now focus on the number density profiles of carbon dioxide and methane inside our pore systems. The first important aspect is that the densities of methane sharply drop in pores that are completely filled with ethaline, levuline, or IL; the local density of CH_4 becomes much higher in pores partially filled with solvent. Such a trend is also observed for the local densities of CO_2 , but the changes in densities of carbon dioxide observed upon variation on the amounts of preadsorbed solvent are not as sharp as those observed for methane, mainly due to the initial CO_2/CH_4 ratio of 5:95. For all graphite systems, CO_2 has its largest local density near the pore walls for both DESs and the IL systems, whereas in rutile systems, significant variations are observed in the local density profiles of CO_2 , depending on the amounts of preadsorbed solvent and the nature of the solvent (ethaline, levuline, IL). As discussed above and in our previous study,⁵¹ the HBD species in the DESs adsorb strongly to the rutile walls due to strong interactions (Figure 8) and depletes other species from the first layers of fluid. Therefore, in all DES systems in titania pores, the largest density of CO_2 is observed farther away from the walls compared to the density peaks of the HBD. In IL systems, strong interactions between CO_2 and the rutile walls (Figure 8) combined with the absence of HBD species lead to large

densities of CO_2 near the titania pore walls, with taller density peaks for carbon dioxide observed in rutile pores partially filled with the IL. As determined in Section 3.3 below, for systems of IL inside pores, the CO_2 –graphite interactions are weaker compared to the CO_2 –rutile energies (Figure 8), and therefore, much smaller densities of CO_2 are observed near the walls in systems IG3 and IG8 as compared to the rutile systems IR3 and IR8. Similar to carbon dioxide, methane has a large local density near the pore walls in all graphite systems, especially in the systems EG3, LG3, and IG3 that have smaller amounts of solvent (Figure 7); in contrast, the peaks in the local density of methane are observed farther away from the walls in rutile systems. Unlike carbon dioxide, methane does not seem to interact strongly with the rutile surfaces (Figure 8) and thus has a smaller density near rutile walls as compared to its density closer to graphite walls; ethaline systems show larger densities of methane near the graphite walls as compared to levuline systems. Large density peaks are observed for methane in the second layers farther away from the rutile walls in the ER3 system, as compared to much smaller peaks in the second layers of the LR3 and IR3. We also note that subtle differences are observed in the local density profiles of carbon dioxide and methane obtained in the simulations using different models for levuline (Figures 7 and S5), although in general, both levuline models considered here yield similar density profiles for both gases.

The local density profiles presented in Figures 7 and S5 showed where carbon dioxide and methane position themselves inside the pores partially and fully filled with DES and IL. In our previous study,⁵¹ where ethaline confined in slit graphitic and rutile pores was put in contact with a bulk gas phase of pure CO_2 at pressures ~ 11 – 18 bar, we found that the average number density of CO_2 dissolved within confined ethaline was smaller than the value found for bulk ethaline. However, if we accounted for all CO_2 inside the nanopores (i.e., dissolved in confined ethaline, adsorbed at the pore walls, adsorbed at the gas/ethaline interface, and present in the gas phase inside the pores), the average number density of CO_2 inside the nanopores was up to 7.3 times of the value obtained for bulk ethaline. Nevertheless, the results shown in Figures 3–5 indicate that pores fully filled with the DESs or IL outperform pores partially filled with solvent. Increases in the amount of solvent inside the pore lead to smaller pore volumes available for the gas phase, which has a CO_2/CH_4 ratio of ~ 5 :95. Therefore, completely filling the pores with the DESs or IL lead to a sharp reduction in the amount of CH_4 present in the pores, as this gas has limited solubility in all solvents considered here. As noted in our discussion of Figures 3–5, gas solubilities alone do not determine the separation performance of our systems, as gas diffusivities also need to be considered. We also note that in rutile systems, the Ti/O atoms at the edges of rutile walls are “hot spots” for adsorption and thus can attract some of the DES or IL components away from inside the pore. This preferential adsorption further decreases the amount of solvent inside the pore volume, which leads to larger pore volumes available for the gas phase. As the average number densities of CO_2 and CH_4 in the gas phase inside the pore are typically larger than the densities of these molecules within the confined liquid (Figures 7 and S5), the movement of liquid solvent to the edges of the rutile walls results in a net increase in the densities of CO_2 and CH_4 inside the pore, as compared to graphitic systems with the same amounts of DES/IL.

3.3. Interaction Energies. Average interaction energies between different entities in our systems (electrostatic + van der Waals) are reported in Figure 8; all relevant data are presented in Tables S5–S7 (Supporting Information). In ethaline systems, the interactions of CO₂ with the HBD and cation are comparable in magnitude, whereas in levuline systems, CO₂ has stronger interactions with the HBD than with the cation. For both DESs, the interactions CO₂–cation and CO₂–HBD are slightly stronger in bulk systems compared to confined systems. The CO₂–HBD interactions in levuline systems are stronger than in ethaline systems; however, the CO₂–cation interactions in levuline systems are weaker than in ethaline systems. The interactions of CO₂ with the graphite walls are stronger than with the rutile walls for both DESs, which is explained by the fact that the ions, but mainly the HBD, interact very strongly with the rutile walls (especially ethylene glycol in ethaline). Therefore, the HBD is the main species adsorbed close to the rutile walls, as seen in the density profiles reported in Figure 7 (especially ethylene glycol), and therefore, very small quantities of all other species are adsorbed next to the rutile walls. In all IL systems, comparable interactions in magnitude are observed between CO₂–cation and CO₂–anion. In bulk systems, these interactions are slightly stronger than the energies observed in graphite pores fully filled with IL. In rutile systems, CO₂ interacts strongly with the titania walls, and thus, it has a large density peak near the pore walls (Figure 7), which leads to weaker interactions with cations and anions compared to what was observed in graphite systems. We note that the average interaction energies computed for graphite pores completely filled with IL are in excellent agreement with the values reported by Budhathoki et al.⁵⁶ Similar trends are observed for the interactions of CH₄ with the other entities in our systems. In general, the interactions between CO₂ and the other entities in our systems are stronger than the interactions between CH₄ and the same entities.

4. CONCLUSIONS

Motivated by the development of SILMs and SILPs, in this study, classical MD simulations were used to investigate the performance of slit graphite and titania (rutile) pores partially and completely filled with DESs or ILs, for gas separations of carbon dioxide from methane. The porous materials have a pore size of 5.2 nm, and the DESs considered were ethaline (an 1:2 molar mixture of choline chloride and ethylene glycol) and levuline (choline chloride and levulinic acid with a molar ratio of 1:2); the IL considered was [bmim⁺][NTf₂[−]]. The performance of these systems in terms of solubility selectivity, diffusion selectivity, and permselectivity to separate a bulk gas mixture of CO₂ and CH₄ with approximate molar ratio of 5:95, at a temperature of 318 K, and pressures on the order of 100 bar was compared against results obtained for carbon and rutile pores without preadsorbed solvent, as well as against the performance of bulk liquid solvents. The latter systems can be also viewed as model systems of the micrometer-sized pores in a supported IL or DES membrane. In terms of solubility selectivity, empty rutile pores have the largest value among all systems evaluated, followed by bulk ethaline, rutile pores completely filled by the IL, graphite and rutile pores filled with ethaline, and bulk levuline. In terms of diffusivity selectivities, the largest values were observed for bulk levuline, followed by ethaline completely filling a rutile nanopore and a graphite nanopore completely filled with the IL; strong adsorption of CO₂ by the titania walls cause the diffusivity selectivity in empty

rutile pores to drop well below unity. When both solubilities and diffusivities are factored into permselectivities, the best performance is obtained for bulk levuline, followed by ethaline fully filling a rutile pore, a graphite pore filled by ethaline, bulk ethaline, and the IL filling a graphite pore. In general, the bulk DESs tend to have larger solubility selectivities as compared to the confined DESs, but the opposite trend is observed for IL systems. Confined ethaline seems to perform better than bulk ethaline in terms of diffusivity selectivity and permselectivity. These observations are driven by the strong adsorption of ethylene glycol by the pore walls, especially in the case of rutile pores, which lead to depletion of the HBD near the center of the pore and might help increase the mobility of CO₂ with respect to CH₄. In contrast, bulk levuline has better diffusivity selectivity and permselectivity compared to confined levuline, as the HBD (levulinic acid) is adsorbed less strongly than ethylene glycol by the rutile or graphite walls. For the IL, confinement in a graphite pore slightly improves its diffusivity selectivity and permselectivity with respect to the bulk IL, but drops in both metrics are observed upon confinement of the IL in a rutile pore. In IL systems within rutile pores, CO₂ was found to interact very strongly with the rutile walls, but in DES systems in rutile pores, the HBD (especially ethylene glycol) is predominantly adsorbed close to the titania walls. In general, systems of nanopores partially filled with solvents are outperformed by the same nanopores fully filled by solvents and by the bulk solvents. Drops of 2–3 orders of magnitude are observed in the gas diffusivity in pores filled with solvents with respect to systems of empty pores, which may be problematic if permeation is mainly controlled by diffusion. However, if adsorption dominates the gas permeation within the membrane, our results suggest that systems of levuline in the micrometer-sized pores of a supported DES membrane or ethaline confined in the rutile nanopores of a supported DES phase material might represent promising systems for gas separation.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.langmuir.8b03990.

Data of solubility, solubility selectivity, diffusivity, diffusivity selectivity, permeabilities, and permselectivity of carbon dioxide and methane; diffusivities for IL and DES species in our systems; data of average interaction energies; additional simulation snapshots; RDFs between different levuline models and discussion; and additional local number density profiles (PDF)

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Notes

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REFERENCES

- (1) Boot-Handford, M. E.; Abanades, J. C.; Anthony, E. J.; Blunt, M. J.; Brandani, S.; Mac Dowell, N.; Fernández, J. R.; Ferrari, M.-C.; Gross, R.; Hallett, J. P.; Haszeldine, R. S.; Heptonstall, P.; Lyngfelt, A.; Makuch, Z.; Mangano, E.; Porter, R. T. J.; Pourkashanian, M.; Rochelle, G. T.; Shah, N.; Yao, J. G.; Fennell, P. S. Carbon capture and storage update. *Energy Environ. Sci.* **2014**, *7*, 130–189.
- (2) Luis, P.; Van Gerven, T.; Van der Bruggen, B. Recent developments in membrane-based technologies for CO₂ capture. *Prog. Energy Combust. Sci.* **2012**, *38*, 419–448.
- (3) D'Alessandro, D. M.; Smit, B.; Long, J. R. Carbon Dioxide Capture: Prospects for New Materials. *Angew. Chem., Int. Ed.* **2010**, *49*, 6058–6082.
- (4) Ramdin, M.; de Loos, T. W.; Vlugt, T. J. H. State-of-the-Art of CO₂ Capture with Ionic Liquids. *Ind. Eng. Chem. Res.* **2012**, *51*, 8149–8177.
- (5) Shiflett, M. B.; Maginn, E. J. The solubility of gases in ionic liquids. *AIChE J.* **2017**, *63*, 4722–4737.
- (6) Bara, J. E.; Carlisle, T. K.; Gabriel, C. J.; Camper, D.; Finotello, A.; Gin, D. L.; Noble, R. D. Guide to CO₂ Separations in Imidazolium-Based Room-Temperature Ionic Liquids. *Ind. Eng. Chem. Res.* **2009**, *48*, 2739–2751.
- (7) Shannon, M. S.; Bara, J. E. Reactive and Reversible Ionic Liquids for CO₂ Capture and Acid Gas Removal. *Sep. Sci. Technol.* **2012**, *47*, 178–188.
- (8) Gutowski, K. E.; Maginn, E. J. Amine-Functionalized Task-Specific Ionic Liquids: A Mechanistic Explanation for the Dramatic Increase in Viscosity upon Complexation with CO₂ from Molecular Simulation. *J. Am. Chem. Soc.* **2008**, *130*, 14690–14704.
- (9) Sheridan, Q. R.; Schneider, W. F.; Maginn, E. J. Role of Molecular Modeling in the Development of CO₂-Reactive Ionic Liquids. *Chem. Rev.* **2018**, *118*, 5242–5260.
- (10) Brennecke, J. F.; Gurkan, B. E. Ionic Liquids for CO₂ Capture and Emission Reduction. *J. Phys. Chem. Lett.* **2010**, *1*, 3459–3464.
- (11) Seo, S.; Quiroz-Guzman, M.; DeSilva, M. A.; Lee, T. B.; Huang, Y.; Goodrich, B. F.; Schneider, W. F.; Brennecke, J. F. Chemically Tunable Ionic Liquids with Aprotic Heterocyclic Anion (AHA) for CO₂ Capture. *J. Phys. Chem. B* **2014**, *118*, 5740–5751.
- (12) Abbott, A. P.; Capper, G.; Davies, D. L.; Rasheed, R. K.; Tambyrajah, V. Novel solvent properties of choline chloride/urea mixtures. *Chem. Commun.* **2002**, 70–71.
- (13) Smith, E. L.; Abbott, A. P.; Ryder, K. S. Deep Eutectic Solvents (DESs) and Their Applications. *Chem. Rev.* **2014**, *114*, 11060–11082.
- (14) García, G.; Aparicio, S.; Ullah, R.; Atilhan, M. Deep Eutectic Solvents: Physicochemical Properties and Gas Separation Applications. *Energy Fuels* **2015**, *29*, 2616–2644.
- (15) Zhang, Q.; De Oliveira Vigier, K.; Royer, S.; Jérôme, F. Deep eutectic solvents: syntheses, properties and applications. *Chem. Soc. Rev.* **2012**, *41*, 7108–7146.
- (16) García, G.; Atilhan, M.; Aparicio, S. Interfacial Properties of Deep Eutectic Solvents Regarding to CO₂ Capture. *J. Phys. Chem. C* **2015**, *119*, 21413–21425.
- (17) Ullah, R.; Atilhan, M.; Anaya, B.; Khraisheh, M.; García, G.; ElKhattat, A.; Tariq, M.; Aparicio, S. A detailed study of cholinium chloride and levulinic acid deep eutectic solvent system for CO₂ capture via experimental and molecular simulation approaches. *Phys. Chem. Chem. Phys.* **2015**, *17*, 20941–20960.
- (18) Sze, L. L.; Pandey, S.; Ravula, S.; Pandey, S.; Zhao, H.; Baker, G. A.; Baker, S. N. Ternary Deep Eutectic Solvents Tasked for Carbon Dioxide Capture. *ACS Sustainable Chem. Eng.* **2014**, *2*, 2117–2123.
- (19) Zubeir, L. F.; Lacroix, M. H. M.; Kroon, M. C. Low transition temperature mixtures as innovative and sustainable CO₂ capture solvents. *J. Phys. Chem. B* **2014**, *118*, 14429–14441.
- (20) Liu, X.; Gao, B.; Jiang, Y.; Ai, N.; Deng, D. Solubilities and Thermodynamic Properties of Carbon Dioxide in Guaiacol-Based Deep Eutectic Solvents. *J. Chem. Eng. Data* **2017**, *62*, 1448–1455.
- (21) Deng, D.; Jiang, Y.; Liu, X.; Zhang, Z.; Ai, N. Investigation of solubilities of carbon dioxide in five levulinic acid-based deep eutectic solvents and their thermodynamic properties. *J. Chem. Thermodyn.* **2016**, *103*, 212–217.
- (22) Ali, E.; Hadj-Kali, M. K.; Mulyono, S.; Alnashef, I.; Fakeeha, A.; Mjalli, F.; Hayyan, A. Solubility of CO₂ in deep eutectic solvents: Experiments and modelling using the Peng-Robinson equation of state. *Chem. Eng. Res. Des.* **2014**, *92*, 1898–1906.
- (23) Chen, Y.; Ai, N.; Li, G.; Shan, H.; Cui, Y.; Deng, D. Solubilities of Carbon Dioxide in Eutectic Mixtures of Choline Chloride and Dihydric Alcohols. *J. Chem. Eng. Data* **2014**, *59*, 1247–1253.
- (24) Mulia, K.; Putri, S.; Krisanti, E.; Nasruddin. Natural deep eutectic solvents (NADES) as green solvents for carbon dioxide capture. *International Conference on Chemistry, Chemical Process and Engineering (IC3PE)*; American Institute of Physics, 2017; p 020022.
- (25) Altamash, T.; Nasser, M. S.; Elhamarnah, Y.; Magzoub, M.; Ullah, R.; Anaya, B.; Aparicio, S.; Atilhan, M. Gas Solubility and Rheological Behavior of Natural Deep Eutectic Solvents (NADES) via Combined Experimental and Molecular Simulation Techniques. *ChemistrySelect* **2017**, *2*, 7278–7295.
- (26) Li, G.; Deng, D.; Chen, Y.; Shan, H.; Ai, N. Solubilities and thermodynamic properties of CO₂ in choline-chloride based deep eutectic solvents. *J. Chem. Thermodyn.* **2014**, *75*, 58–62.
- (27) Leron, R. B.; Li, M.-H. Solubility of carbon dioxide in a choline chloride-ethylene glycol based deep eutectic solvent. *Thermochim. Acta* **2013**, *551*, 14–19.
- (28) Leron, R. B.; Li, M.-H. Solubility of carbon dioxide in a eutectic mixture of choline chloride and glycerol at moderate pressures. *J. Chem. Thermodyn.* **2013**, *57*, 131–136.
- (29) Mirza, N. R.; Nicholas, N. J.; Wu, Y.; Mumford, K. A.; Kentish, S. E.; Stevens, G. W. Experiments and Thermodynamic Modeling of the Solubility of Carbon Dioxide in Three Different Deep Eutectic Solvents (DESs). *J. Chem. Eng. Data* **2015**, *60*, 3246–3252.
- (30) Lu, M.; Han, G.; Jiang, Y.; Zhang, X.; Deng, D.; Ai, N. Solubilities of carbon dioxide in the eutectic mixture of levulinic acid (or furfuryl alcohol) and choline chloride. *J. Chem. Thermodyn.* **2015**, *88*, 72–77.
- (31) Francisco, M.; van den Bruinhorst, A.; Zubeir, L. F.; Peters, C. J.; Kroon, M. C. A new low transition temperature mixture (LTTM) formed by choline chloride+lactic acid: Characterization as solvent for CO₂ capture. *Fluid Phase Equilib.* **2013**, *340*, 77–84.
- (32) Leron, R. B.; Caparanga, A.; Li, M.-H. Carbon dioxide solubility in a deep eutectic solvent based on choline chloride and urea at T=303.15–343.15K and moderate pressures. *J. Taiwan Inst. Chem. Eng.* **2013**, *44*, 879–885.
- (33) Li, X.; Hou, M.; Han, B.; Wang, X.; Zou, L. Solubility of CO₂ in a Choline Chloride + Urea Eutectic Mixture. *J. Chem. Eng. Data* **2008**, *53*, 548–550.
- (34) Aissaoui, T.; AlNashef, I. M.; Qureshi, U. A.; Benguerba, Y. Potential applications of deep eutectic solvents in natural gas sweetening for CO₂ capture. *Rev. Chem. Eng.* **2017**, *33*, 523–550.
- (35) Sarmad, S.; Mikkola, J.-P.; Ji, X. Carbon Dioxide Capture with Ionic Liquids and Deep Eutectic Solvents: A New Generation of Sorbents. *ChemSusChem* **2016**, *10*, 324–352.
- (36) Kenarsari, S. D.; Yang, D.; Jiang, G.; Zhang, S.; Wang, J.; Russell, A. G.; Wei, Q.; Fan, M. Review of recent advances in carbon dioxide separation and capture. *RSC Adv.* **2013**, *3*, 22739–22773.
- (37) Samanta, A.; Zhao, A.; Shimizu, G. K. H.; Sarkar, P.; Gupta, R. Post-Combustion CO₂ Capture Using Solid Sorbents: A Review. *Ind. Eng. Chem. Res.* **2011**, *51*, 1438–1463.
- (38) Salar-García, M. J.; Ortiz-Martínez, V. M.; Hernández-Fernández, F. J.; de los Ríos, A. P.; Quesada-Medina, J. Ionic liquid

technology to recover volatile organic compounds (VOCs). *J. Hazard. Mater.* **2017**, *321*, 484–499.

(39) Tomé, L. C.; Marrucho, I. M. Ionic liquid-based materials: a platform to design engineered CO₂ separation membranes. *Chem. Soc. Rev.* **2016**, *45*, 2785–2824.

(40) Dai, Z.; Noble, R. D.; Gin, D. L.; Zhang, X.; Deng, L. Combination of ionic liquids with membrane technology: A new approach for CO₂ separation. *J. Membr. Sci.* **2016**, *497*, 1–20.

(41) Duan, Y.; Wu, Y.; Yu, S.; Li, D. Separation of CO₂ with Supported Ionic Liquid Membrane. *Prog. Chem.* **2012**, *24*, 1405–1412.

(42) Liu, Y.; Ma, Y.; Xu, Q.; Yin, J. Supported Ionic Liquid Membrane: Preparation, Characterization and Stability. *Prog. Chem.* **2013**, *25*, 1795–1804.

(43) Lozano, L. J.; Godínez, C.; de los Ríos, A. P.; Hernández-Fernández, F. J.; Sánchez-Segado, S.; Alguacil, F. J. Recent advances in supported ionic liquid membrane technology. *J. Membr. Sci.* **2011**, *376*, 1–14.

(44) Perdikaki, A. V.; Vangeli, O. C.; Karanikolos, G. N.; Stefanopoulos, K. L.; Beltsios, K. G.; Alexandridis, P.; Kanellopoulos, N. K.; Romanos, G. E. Ionic Liquid-Modified Porous Materials for Gas Separation and Heterogeneous Catalysis. *J. Phys. Chem. C* **2012**, *116*, 16398–16411.

(45) Kohler, F. T. U.; Popp, S.; Klefer, H.; Eckle, I.; Schrage, C.; Böhringer, B.; Roth, D.; Haumann, M.; Wasserscheid, P. Supported ionic liquid phase (SILP) materials for removal of hazardous gas compounds—efficient and irreversible NH₃ adsorption. *Green Chem.* **2014**, *16*, 3560–3568.

(46) Bara, J. E.; Camper, D. E.; Gin, D. L.; Noble, R. D. Room-Temperature Ionic Liquids and Composite Materials: Platform Technologies for CO₂ Capture. *Acc. Chem. Res.* **2010**, *43*, 152–159.

(47) Abedini, A.; Crabtree, E.; Bara, J. E.; Turner, C. H. Molecular analysis of selective gas adsorption within composites of ionic polyimides and ionic liquids as gas separation membranes. *Chem. Phys.* **2019**, *516*, 71–83.

(48) Cheng, L.-H.; Rahaman, M. S. A.; Yao, R.; Zhang, L.; Xu, X.-H.; Chen, H.-L.; Lai, J.-Y.; Tung, K.-L. Study on microporous supported ionic liquid membranes for carbon dioxide capture. *Int. J. Greenhouse Gas Control* **2014**, *21*, 82–90.

(49) Althuluth, M.; Overbeek, J. P.; van Wees, H. J.; Zubeir, L. F.; Haije, W. G.; Berrouk, A.; Peters, C. J.; Kroon, M. C. Natural gas purification using supported ionic liquid membrane. *J. Membr. Sci.* **2015**, *484*, 80–86.

(50) Zhang, S.; Zhang, J.; Zhang, Y.; Deng, Y. Nanoconfined Ionic Liquids. *Chem. Rev.* **2016**, *117*, 6755–6833.

(51) Shen, Y.; Hung, F. R. A Molecular Simulation Study of Carbon Dioxide Uptake by a Deep Eutectic Solvent Confined in Slit Nanopores. *J. Phys. Chem. C* **2017**, *121*, 24562–24575.

(52) García, G.; Atilhan, M.; Aparicio, S. The impact of charges in force field parameterization for molecular dynamics simulations of deep eutectic solvents. *J. Mol. Liq.* **2015**, *211*, 506–514.

(53) Mainberger, S.; Kindlein, M.; Bezold, F.; Elts, E.; Minceva, M.; Briesen, H. Deep eutectic solvent formation: a structural view using molecular dynamics simulations with classical force fields. *Mol. Phys.* **2017**, *115*, 1309–1321.

(54) Hazelbaker, E. D.; Budhathoki, S.; Katihar, A.; Shah, J. K.; Maginn, E. J.; Vasenkov, S. Combined Application of High-Field Diffusion NMR and Molecular Dynamics Simulations To Study Dynamics in a Mixture of Carbon Dioxide and an Imidazolium-Based Ionic Liquid. *J. Phys. Chem. B* **2012**, *116*, 9141–9151.

(55) Singh, R.; Marin-Rimoldi, E.; Maginn, E. J. A Monte Carlo Simulation Study To Predict the Solubility of Carbon Dioxide, Hydrogen, and Their Mixture in the Ionic Liquids 1-Alkyl-3-methylimidazolium bis(trifluoromethanesulfonyl)amide ([Cnmim+][Tf₂N⁻], n = 4, 6). *Ind. Eng. Chem. Res.* **2014**, *54*, 4385–4395.

(56) Budhathoki, S.; Shah, J. K.; Maginn, E. J. Molecular Simulation Study of the Performance of Supported Ionic Liquid Phase Materials for the Separation of Carbon Dioxide from Methane and Hydrogen. *Ind. Eng. Chem. Res.* **2017**, *56*, 6775–6784.

(57) Rajput, N. N.; Monk, J.; Singh, R.; Hung, F. R. On the Influence of Pore Size and Pore Loading on Structural and Dynamical Heterogeneities of an Ionic Liquid Confined in a Slit Nanopore. *J. Phys. Chem. C* **2012**, *116*, 5169–5181.

(58) Rajput, N. N.; Monk, J.; Hung, F. R. Structure and Dynamics of an Ionic Liquid Confined Inside a Charged Slit Graphitic Nanopore. *J. Phys. Chem. C* **2012**, *116*, 14504–14513.

(59) Singh, R.; Rajput, N. N.; He, X.; Monk, J.; Hung, F. R. Molecular dynamics simulations of the ionic liquid [EMIM+][TFMSI⁻] confined inside rutile (110) slit nanopores. *Phys. Chem. Chem. Phys.* **2013**, *15*, 16090–16103.

(60) Shen, Y.; He, X.; Hung, F. R. Structural and Dynamical Properties of a Deep Eutectic Solvent Confined Inside a Slit Pore. *J. Phys. Chem. C* **2015**, *119*, 24489–24500.

(61) Berendsen, H. J. C.; van der Spoel, D.; van Drunen, R. GROMACS: A message-passing parallel molecular dynamics implementation. *Comput. Phys. Commun.* **1995**, *91*, 43–56.

(62) Lindahl, E.; Hess, B.; van der Spoel, D. GROMACS 3.0: a package for molecular simulation and trajectory analysis. *J. Mol. Model.* **2001**, *7*, 306–317.

(63) Van Der Spoel, D.; Lindahl, E.; Hess, B.; Groenhof, G.; Mark, A. E.; Berendsen, H. J. C. GROMACS: fast, flexible, and free. *J. Comput. Chem.* **2005**, *26*, 1701–1718.

(64) Hess, B.; Kutzner, C.; van der Spoel, D.; Lindahl, E. GROMACS 4: Algorithms for Highly Efficient, Load-Balanced, and Scalable Molecular Simulation. *J. Chem. Theory Comput.* **2008**, *4*, 435–447.

(65) Pronk, S.; Páll, S.; Schulz, R.; Larsson, P.; Bjelkmar, P.; Apostolov, R.; Shirts, M. R.; Smith, J. C.; Kasson, P. M.; van der Spoel, D.; Hess, B.; Lindahl, E. GROMACS 4.5: a high-throughput and highly parallel open source molecular simulation toolkit. *Bioinformatics* **2013**, *29*, 845–854.

(66) Páll, S.; Abraham, M. J.; Kutzner, C.; Hess, B.; Lindahl, E. Tackling exascale software challenges in molecular dynamics simulations with GROMACS. *International Conference on Exascale Applications and Software*; Springer, 2014; pp 3–27.

(67) Abraham, M. J.; Murtola, T.; Schulz, R.; Páll, S.; Smith, J. C.; Hess, B.; Lindahl, E. GROMACS: High performance molecular simulations through multi-level parallelism from laptops to supercomputers. *SoftwareX* **2015**, *1–2*, 19–25.

(68) Schüttelkopf, A. W.; van Aalten, D. M. F. PRODRG: a tool for high-throughput crystallography of protein-ligand complexes. *Acta Crystallogr., Sect. D: Biol. Crystallogr.* **2004**, *60*, 1355–1363.

(69) Potoff, J. J.; Siepmann, J. I. Vapor-liquid equilibria of mixtures containing alkanes, carbon dioxide, and nitrogen. *AIChE J.* **2001**, *47*, 1676–1682.

(70) Perkins, S. L.; Painter, P.; Colina, C. M. Experimental and Computational Studies of Choline Chloride-Based Deep Eutectic Solvents. *J. Chem. Eng. Data* **2014**, *59*, 3652–3662.

(71) Wang, J.; Wolf, R. M.; Caldwell, J. W.; Kollman, P. A.; Case, D. A. Development and testing of a general amber force field. *J. Comput. Chem.* **2004**, *25*, 1157–1174.

(72) Perkins, S. L.; Painter, P.; Colina, C. M. Molecular dynamic simulations and vibrational analysis of an ionic liquid analogue. *J. Phys. Chem. B* **2013**, *117*, 10250–10260.

(73) Frisch, M. J.; Trucks, G. W.; Schlegel, H. B.; Scuseria, G. E.; Robb, M. A.; Cheeseman, J. R.; Scalmani, G.; Barone, V.; Petersson, G. A.; Nakatsuji, H.; Li, X.; Caricato, M.; Marenich, A.; Bloino, J.; Janesko, B. G.; Gomperts, R.; Mennucci, B.; Hratchian, H. P.; Ortiz, J. V.; Izmaylov, A. F.; Sonnenberg, J. L.; Williams-Young, D.; Ding, F.; Lipparini, F.; Egidi, F.; Goings, J.; Peng, B.; Petrone, A.; Henderson, T.; Ranasinghe, D.; Zakrzewski, V. G.; Gao, J.; Rega, N.; Zheng, G.; Liang, W.; Hada, M.; Ehara, M.; Toyota, K.; Fukuda, R.; Hasegawa, J.; Ishida, M.; Nakajima, T.; Honda, Y.; Kitao, O.; Nakai, H.; Vreven, T.; Throssell, K.; Montgomery, J. A., Jr.; Peralta, J. E.; Ogliaro, F.; Bearpark, M.; Heyd, J. J.; Brothers, E.; Kudin, K. N.; Staroverov, V. N.; Keith, T.; Kobayashi, R.; Normand, J.; Raghavachari, K.; Rendell, A.; Burant, J. C.; Iyengar, S. S.; Tomasi, J.; Cossi, M.; Millam, J. M.; Klene, M.; Adamo, C.; Cammi, R.; Ochterski, J. W.; Martin, R. L.; Morokuma,

K.; Farkas, O.; Foresman, J. B.; Fox, D. J. *Gaussian 16*, Revision A.03; Gaussian, Inc.: Wallingford CT, 2016.

(74) Case, D. A.; Ben-Shalom, I. Y.; Brozell, S. R.; Cerutti, D. S.; Cheatham, T. E. I.; Cruzeiro, V. W. D.; Darden, T. A.; Duke, R. E.; Ghoreishi, D.; Gilson, M. K.; Gohlke, H.; Goetz, A. W.; Greene, D.; Harris, R.; Homeyer, N.; Izadi, S.; Kovalenko, A.; Kurtzman, T.; Lee, T. S.; LeGrand, S.; Li, P.; Lin, C.; Liu, J.; Luchko, T.; Luo, R.; Mermelstein, D. J.; Merz, K. M.; Miao, Y.; Monard, G.; Nguyen, C.; Nguyen, H.; Omelyan, I.; Onufriev, A.; Pan, F.; Qi, R.; Roe, D. R.; Roitberg, A.; Sagui, C.; Schott-Verdugo, S.; Shen, J.; Simmerling, C. L.; Smith, J.; Salomon-Ferrer, R.; Swails, J.; Walker, R. C.; Wang, J.; Wei, H.; Wolf, R. M.; Wu, X.; Xiao, L.; York, D. M.; Kollman, P. A. *AMBER 2018*; University of California: San Francisco, 2018.

(75) Bayly, C. L.; Cieplak, P.; Cornell, W.; Kollman, P. A. A well-behaved electrostatic potential based method using charge restraints for deriving atomic charges: the RESP model. *J. Phys. Chem.* **1993**, *97*, 10269–10280.

(76) Abedin, R.; Heidarian, S.; Flake, J. C.; Hung, F. R. Computational Evaluation of Mixtures of Hydrofluorocarbons and Deep Eutectic Solvents for Absorption Refrigeration Systems. *Langmuir* **2017**, *33*, 11611–11625.

(77) Martínez, L.; Andrade, R.; Birgin, E. G.; Martínez, J. M. PACKMOL: a package for building initial configurations for molecular dynamics simulations. *J. Comput. Chem.* **2009**, *30*, 2157–2164.

(78) Neves, L. A.; Crespo, J. G.; Coelho, I. M. Gas permeation studies in supported ionic liquid membranes. *J. Membr. Sci.* **2010**, *357*, 160–170.

(79) Avdibegović, D.; Regadío, M.; Binnemans, K. Recovery of scandium(III) from diluted aqueous solutions by a supported ionic liquid phase (SILP). *RSC Adv.* **2017**, *7*, 49664–49674.

(80) Vegt, N. F. A. v. d.; Briels, W. J.; Wessling, M.; Strathmann, H. Free energy calculations of small molecules in dense amorphous polymers. Effect of the initial guess configuration in molecular dynamics studies. *J. Chem. Phys.* **1996**, *105*, 8849–8857.

(81) Rallabandi, P. S.; Ford, D. M. Permeation of small molecules through polymers confined in mesoporous media. *J. Membr. Sci.* **2000**, *171*, 239–252.

(82) Elliott, J. R.; Lira, C. T., *Introductory Chemical Engineering Thermodynamics*, 2nd ed.; Prentice Hall: Upper Saddle River, NJ, 2012.

(83) Reid, R. C.; Prausnitz, J. M.; Poling, B. E. *The Properties of Gases and Liquids*, 4th ed.; McGraw-Hill: New York, 1987. p 83.

(84) Fujie, K.; Kitagawa, H. Ionic liquid transported into metal-organic frameworks. *Coord. Chem. Rev.* **2016**, *307*, 382–390.

(85) Martínez-Palou, R.; Likhanova, N. V.; Olivares-Xometl, O. Supported ionic liquid membranes for separations of gases and liquids: an overview. *Pet. Chem.* **2015**, *54*, 595–607.

(86) Towns, J.; Cockerill, T.; Dahan, M.; Foster, I.; Gaither, K.; Grimshaw, A.; Hazlewood, V.; Lathrop, S.; Lifka, D.; Peterson, G. D.; Roskies, R.; Scott, J. R.; Wilkens-Diehr, N. XSEDE: Accelerating Scientific Discovery. *Comput. Sci. Eng.* **2014**, *16*, 62–74.